Investment and Financial Management

Financial Markets

Interests

Agenda: Interests

Core Principle I: The Time Value of Money

I	It is only possible to **<u>compare</u>** or **<u>combine</u>** values that refer to the **same point in time**.
II	To move a cash flow forward in time, you must **compound** it (Future Value).
III	To move a cash flow back in time, we must **discount** it (Present Value).

The interest rate is applied to convert time values from one period to another!

Terms & Definitions

Interest as the "price of money" which has to be paid for borrowing money

In exchange for the use of money, a debtor/borrower pays a fraction of the account balance back to the creditor/lender.

Creditor/ Lender

Capital (C_0) →

← Capital + Interest (C_N)

Debtor/ Borrower

C_0	Initial cash flow at t_0 (start)
C_n	Cash flow at date n
N	Date of last cash flow in a stream of cash flows
r	Interest rate or discount rate

Different kinds of interest rates

Which different kinds of interest rates do we know?

Interest rate	Characteristic
Deposit rate	Received for deposits at a bank
Debt (borrowing) rate	Paid for borrowing capital from the bank
Prime rate	Interest rate at which banks lend to their most creditworthy clients
Key interest rates or federal funds rate	Interest rates at which central banks borrow funds from or lend funds to commercial banks (ECB: see next slide; FED: target federal funds rate)
Money market rates	(Risk-free) Interest rate on short-term (≈ up to 1 year) transactions
Interbank rates	Rates at which banks lend to each other
Bond yields	Yield that can be earned on mid-term and long-term (risk-free) financial instruments
Nominal rate	Interest rate fixed in financial contracts
Effective rate	Effective interest rate paid/earned in a financial contract
Real interest rate	Yield earned above inflation rate

ECB Key interest rates

Interest rates which apply to banks' interaction with the central bank

The key interest rates can be subdivided into …

i. **Deposit facility:** interest on deposit of banks at the central bank which can become negative

ii. **Main refinancing rate:** publicly visible base rate announced by the central bank

iii. **Marginal lending rate:** allows financial institutions to borrow money from the central bank on a short-term basis, e.g. overnight

ECB Key interest rates over time

Date (with effect from)		Deposit facility	Main refinancing operations		Marginal lending facility
			Fixed rate tenders Fixed rate	Variable rate tenders Minimum bid rate	
2023	20 Sep.	4.00	4.50	-	4.75
2023	2 Aug.	3.75	4.25	-	4.50
2023	21 Jun.	3.50	4.00	-	4.25
2023	10 May	3.25	3.75	-	4.00
2023	22 Mar.	3.00	3.50	-	3.75
2023	8 Feb.	2.50	3.00	-	3.25
2022	21 Dec.	2.00	2.50	-	2.75
2022	2 Nov.	1.50	2.00	-	2.25
2022	14 Sep.	0.75	1.25	-	1.50
2022	27 Jul.	0.00	0.50	-	0.75
2019	18 Sep.	−0.50	0.00	-	0.25

Source: ECB

Benchmark interest rates (RFRs)

	United States	United Kingdom	Euro area	Switzerland	Japan
Alternative rate	**SOFR** (secured overnight financing rate)	**SONIA** (sterling overnight index average)	**ESTER** (euro short-term rate)	**SARON** (Swiss average overnight rate)	**TONA** (Tokyo overnight average rate)
Administrator	Federal Reserve Bank of New York	Bank of England	ECB	SIX Swiss Exchange	Bank of Japan
Data source	Triparty repo, FICC GCF, FICC bilateral	Form SMMD (BoE data collection)	MMSR	CHF interbank repo	Money market brokers
Wholesale non-bank counterparties	**Yes**	**Yes**	**Yes**	**No**	**Yes**
Secured	**Yes**	**No**	**No**	**Yes**	**No**
Overnight rate	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
Available now?	Yes	Yes	Oct 2019	Yes	Yes

FICC = Fixed Income Clearing Corporation; GCF = general collateral financing; MMSR = money market statistical reporting; SMMD = sterling money market data collection reporting.

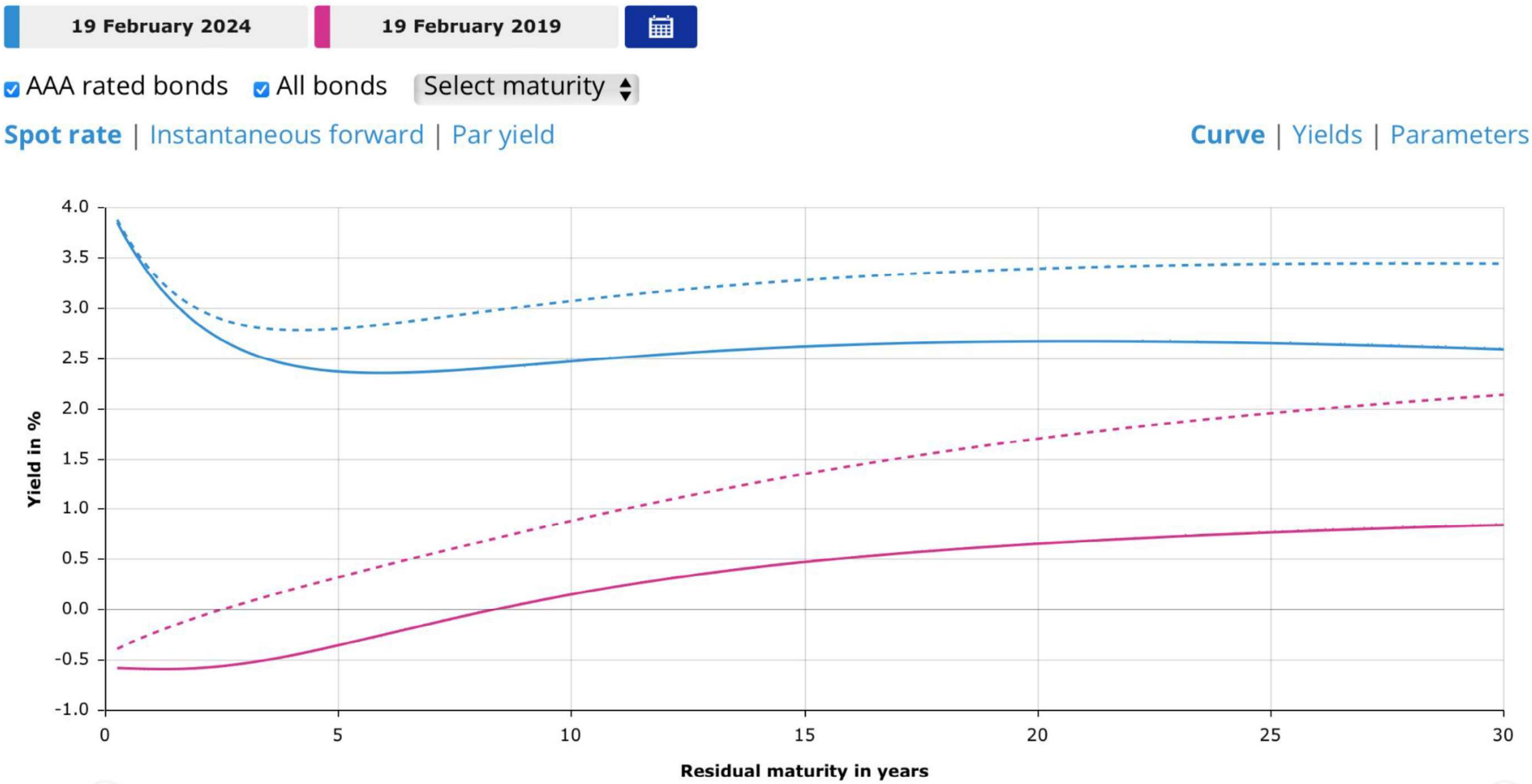
Source: Schrimpf/Sushko (2019), BIS

- After evidence on LIBOR rigging has emerged, it was decided to phase-out the LIBOR as a benchmark interest rate.
- EURIBOR (published by EMMI at 11 am for 5 tenors by using a panel of 20 large banks) is still in use and important because of its role in derivatives markets

Which financial instruments are based on interests?

Financial security	**Characteristic**
Current (checking) account	Used for payment purposes with no or only very low interest
Saving account	Investment with unlimited duration, no payment purposes
Money market instruments	Treasury Bills, Bubills, Federal Treasury Notes, short-term government bonds, commercial papers, etc.
Bonds	Interest-bearing security (usually with fixed maturity) such as zero bond, coupon bond, government bond, convertible, floating rate note
Loan or credit	Banks entrust money to the debtor against interest payment (mortgage loan, working capital loan)
Money market funds	Investment funds investing in money market instruments so that the money market rate can be earned

Euro area term structure



Source: ECB

Agenda: Interests

Simple interest calculation

- One-time interest payment **without compounding effect**
  - Interest is paid on a certain initial amount C_0 which was paid in at the beginning of the investment period
  - Interest calculation is based on fixed interest rate r
  - Only capital which is paid in at the beginning of the investment period C_0 is considered for interest calculations

- **Account balance after n years**

$$(1):\ C_1 = C_0 + C_0 * \mathrm{r} = C_0 * (1 + \mathrm{r})$$

$$(2):\ C_2 = C_1 + C_0 * r = C_0 * (1 + 2 * \mathrm{r})$$

$$(3):\ C_n = C_{n-1} + C_0 * r = C_0 * (1 + \mathrm{n} * r)$$

Simple interest - Example

(A.1) Simple interest will be paid on an initial amount $C_0 = 20{,}000\ EUR$ with an interest rate of $r = 5.5\%$. What will be the payoff after three years? Please provide the calculation formula.

$A = 20{,}000\ EUR$	$B = 20{,}550\ EUR$	$C = 21{,}100\ EUR$	$D = 23{,}300\ EUR$

Present value

Present Value (PV)

- **Reverse idea**: Amount of money one needs to invest at a given interest rate to receive a certain payment C_n at the end of the investment period.
- Calculation of the Present Value by discounting all future cash flows with the given interest rate r
- Example with fixed interest rate and no compounding

$$(1): C_0 * (1 + n * r) = C_n$$

$$(2): C_0 = \frac{C_n}{(1 + n * r)}$$

Present value - Example

(A.2) What is the amount you would need to invest at a fixed interest rate of $r = 8.00\%$ p.a. without compounding in order to receive a payoff of 5,000 EUR in two years?

Simple interest with varying time periods

- How do we deal with transactions or securities with an investment period of less than a year? How do we determine the exact investment period?

- **Several day count conventions**
  - **30/360** (German method applied for German saving accounts): Every month is considered to have 30 days and every year to have 360 days.
  - **Actual/360** (French method applied in money markets for short-term lending of currencies): Every month is considered to have the actual number of days, a year is considered to have 360 days.
  - **Actual/Actual** (applied in EURO/U.S. bond markets): Every month and every year are considered to have the actual number of days. Its also called the ICMA rule.

Simple interest with varying time periods

Simple interest with varying time periods

How do we need to adjust the interest calculation when the actual investment period equals a fraction of the predetermined investment period?

Replace the investment period n by the fraction f of the predetermined period:

$$(1): C_f = C_0 * (1 + f * r)$$

$$(2):\ with\ f = \frac{Interest\ days\ of\ the\ investment}{Interest\ days\ per\ year}$$

Simple interest with varying time periods

(A.3) An amount of 7,500 EUR was invested at a fixed interest rate of 2.85% between May 15 and September 16 of the same year. What would be the payoff at the end of the investment period if you apply the 30/360 method?

Simple interest with varying time periods

(A.4) What amount of money would you need to put on a saving account with a fixed interest rate of 2.85% on 14 April 2018 to be able to buy your favorite car on 23 December 2018 for 22,500 EUR? Please apply the 30/360 method!

Agenda: Interests

Compounding interests: The effect of earning "interest on interest"

- Interest payments are made at the end of the interest period
- Assumption: Immediate reinvestment of interest payment at the beginning of the next interest period

Account balance after n years

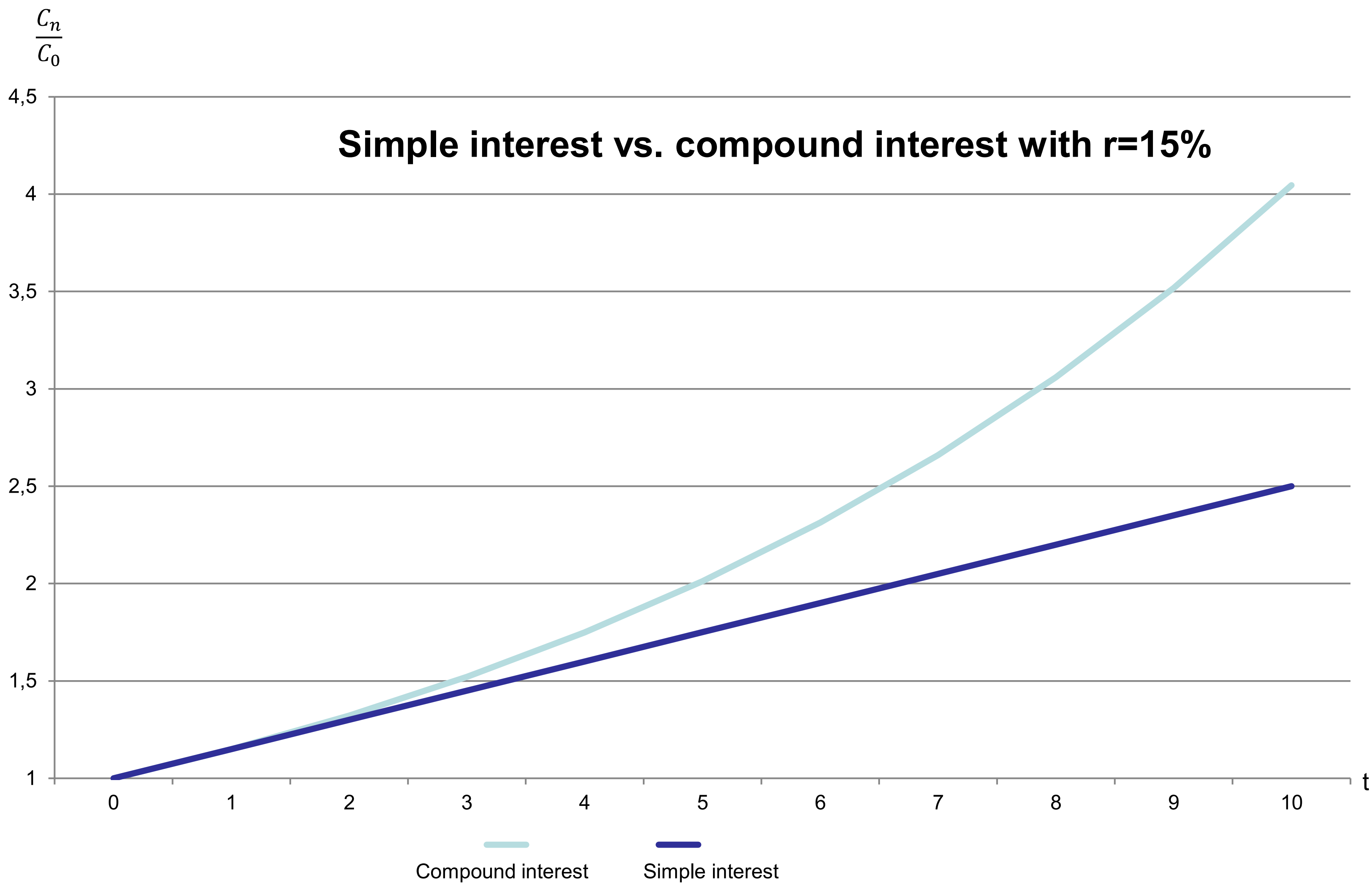
$(1)\!: C_1 = C_0 * (1 + \mathrm{r}) = C_0 * (1 + \mathrm{r})$

$(2)\!: \mathrm{Let's\ define:}\ (1 + \mathrm{r}) = \mathrm{q}$

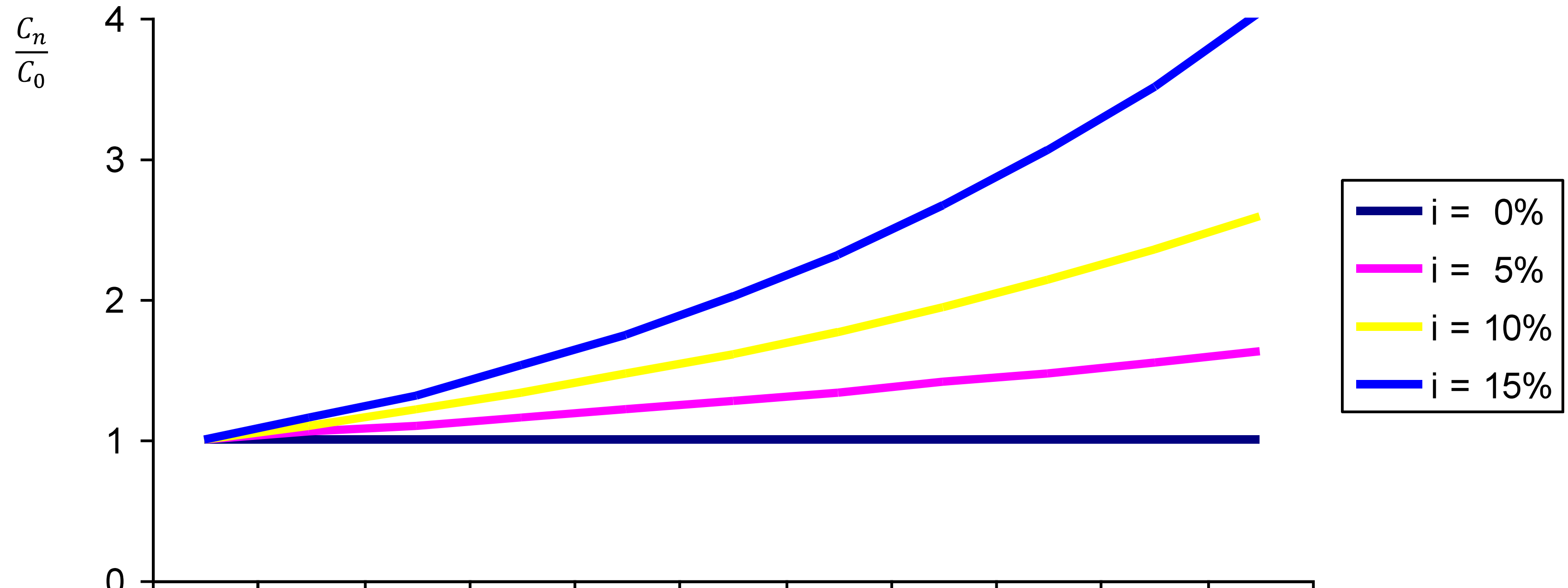
$(3)\!: C_2 = C_1 * (1 + r) = C_0 * (1 + \mathrm{r})^2 = C_0 * q^2$

$(4)\!: C_n = C_{n-1} * (1 + r) = C_0 * (1 + r)^n = C_0 * q^n$

The power of compounding



The power of compounding



Interest periods	0	1	2	3	4	5	6	7	8	9	10
i = 0%	1	1	1	1	1	1	1	1	1	1	1
i = 5%	1	1.05	1.103	1.158	1.216	1.276	1.34	1.407	1.477	1.551	1.629
i = 10%	1	1.1	1.21	1.331	1.464	1.611	1.772	1.949	2.144	2.358	2.594
i = 15%	1	1.15	1.323	1.521	1.749	2.011	2.313	2.66	3.059	3.518	4.046

Compounding Interest - Formula

1. Future Value

$$C_n = C_0 * (1+r)^n = C_0 * q^n$$

2. Present Value

$$C_0 = \frac{C_n}{(1+r)^n} = \frac{C_n}{q^n}$$

3. Time

$$q^n = \frac{C_n}{C_0}; \ln(q^n) = \ln\left(\frac{C_n}{C_0}\right)$$

$$n * \ln(q) = \ln(C_n) - \ln(C_0)$$

$$n = \frac{\ln(C_n) - \ln(C_0)}{\ln(q)}$$

4. Interest Rate

$$q = \sqrt[n]{\frac{C_n}{C_0}}$$

$$r = \sqrt[n]{\frac{C_n}{C_0}} - 1$$

(A.5a) What is the interest rate p.a. which has to be applied to an investment C_0 so that the amount doubles within ten years if we assume compounding interest calculation? Which rate would provide a repayment of 1.5 times the initial amount?

Compounding Interest - Example

(A.5b) How long does it take for an initial investment C_0 to double if the interest rate equals 5% p.a. if we assume compounding interest payments on an annually basis?